HW 6

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(a) Let $D^2 = \min_{2 \le i \le n} \|X_1 - X_i\|^2$. For $0 \le t \le 1$, define a disk $B(x, \sqrt{t}) = \{y : \|y - x\|_2 \le \sqrt{t}\}$. Then, even in the worse case, when x is at the corner of the unit square, at least one quarter of the disk lies inside the unit square. Therefore, $\text{Area}(B(x, \sqrt{t}) \cap [0,1]^2) \ge \frac{1}{4}\pi(\sqrt{t})^2 = \frac{\pi t}{4}$. Since $X_2, \cdots, X_n$ are independent and uniformly distributed in $[0,1]^2$, we have:

$$\begin{aligned} P(D^2 > t) &= P(X_i \notin B(x, \sqrt{t}) \text{ for all } i = 2, \cdots, n | X_1 = x) \\ &= [1 - \text{Area}(B(x, \sqrt{t}) \cap [0,1]^2)]^{n-1} \\ &\le \left(1 - \frac{\pi t}{4}\right)^{n-1} \end{aligned}$$

Use the fact that $1 - x \le e^{-x}$ for all $x \in \mathbb{R}$, we have $P(D^2 > t | X_1 = x) \le e^{-(n-1)\pi t/4}$. This bound doesn't actually depend on x, so $P(D^2 > t) \le e^{-(n-1)\pi t/4}$ for $0 \le t \le 1$. Using the tail integral identity for a nonnegative random variable X, we have $E[D^2] = \int_0^\infty P(D^2 > t)dt$. Since any two points in the unit square are at most $\sqrt{2}$ apart, $D^2 \le 2$, so we can bound the integral as follows: $E[D^2] = \int_0^1 P(D^2 > t)dt + \int_1^2 P(D^2 > t)dt$. We can simplify the first term:

$$\begin{aligned} \int_0^1 P(D^2 > t)dt &\le \int_0^1 e^{-(n-1)\pi t/4} dt \\ &\le \int_0^\infty e^{-(n-1)\pi t/4} dt \\ &= \frac{4}{(n-1)\pi} \end{aligned}$$

For $1 \le t \le 2$, $P(D^2 > t) \le P(D^2 > 1) \le e^{-(n-1)\pi/4}$, so we can bound the second term simply at $e^{-(n-1)\pi/4}$. Therefore, we have $E[D^2] \le \frac{4}{(n-1)\pi} + e^{-(n-1)\pi/4}$, which is $O(1/n)$, since for $n \ge 2$, $\frac{1}{n-1} \le \frac{2}{n}$ and exponential decay is faster than $1/n$. Thus, there exists some absolute constant $C > 0$ such that $E[D^2] \le C/n$ for $n \ge 2$.

(b) Let $Z = f(X_1, \cdots, X_n)$, the total length of the shortest traveling salesman path through all n points. We use the Efron-Stein inequality $Var(Z) \leq \sum_{i=1}^{n} EVar^{(i)}(Z)$. For each i, let $Z_{-i} = f(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n)$, the optimal path length after deleting X_i. Define the nearest neighbor distance of X_i among the other points as $d_i = \min_{j \neq i} \|X_i - X_j\|_2$. We first show that $0 \leq Z - Z_{-i} \leq 2d_i$. The lower bound is trivial, since removing a point cannot increase the optimal path length. For the upper bound, let X_j be the nearest neighbor of X_i so that their distance is d_i. Start with an optimal path through $n-1$ points excluding X_i, then insert X_i between X_j and X_k, its nearest neighbor. The increase in path length is consequently $\|X_j - X_i\|_2 + \|X_i - X_k\|_2 - \|X_j - X_k\|_2$. By the triangle inequality, $\|X_j - X_k\|_2 \leq \|X_j - X_i\|_2 + \|X_i - X_k\|_2$, so the increase in path length is at most $2\|X_j - X_i\|_2 = 2d_i$. Therefore, $0 \leq Z - Z_{-i} \leq 2d_i$.

For fixed X_{-i}, Z_{-i} is constant with respect to X_i. For any random variable Y and constant a, $Var(Y) = E[(Y-a)^2] - (E[Y]-a)^2 \leq E[(Y-a)^2]$. since the mean minimizes the mean squared error. Therefore, conditionally on X_{-i}, we have $Var^{(i)}(Z) \leq E^{(i)}[(Z - Z_{-i})^2] \leq E^{(i)}[4d_i^2] = 4E^{(i)}[d_i^2]$. Taking the expectation over the remaining variables, we have $EVar^{(i)}(Z) \leq 4E[d_i^2]$. By Efron-Stein, $Var(Z) \leq \sum_{i=1}^{n} EVar^{(i)}(Z) \leq 4\sum_{i=1}^{n} E[d_i^2]$. Since all X_i are iid, all d_i follow the same distribution, so $Var(Z) \leq 4nE[d_1^2]$. By part (a), $Var(Z) \leq 4nE[d_1^2] = O(1)$, since $E[d_1^2] = O(1/n)$, thus $Var(Z) \leq C$ for some absolute constant $C > 0$.

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Let $X \sim \text{Exp}(1)$ and $g : \mathbb{R} \to \mathbb{R}$ be bounded with a bounded derivative. Define $h(x) = g(x) - g(0)$, so that $h(0) = 0$ and $h'(x) = g'(x)$. Since variance is unchanged by subtracting a constant, $Var(g(X)) = Var(h(X))$, then $Var(h(X)) = E[h(X)^2] - (E[h(X)])^2 \leq E[h(X)^2]$. Now, using the hint and since $(h(x)^2)' = 2h(x)h'(x)$, we have:

$$\begin{aligned} E[h(X)^2] &= \int_0^\infty h(x)^2 e^{-x} dx \\ &= h(0)^2 + \int_0^\infty 2h(x)h'(x)e^{-x} dx \\ &= 2E[h(X)h'(X)] \end{aligned}$$

Then, by Cauchy-Schwarz

$$\begin{aligned}
E[h(X)h'(X)] &\leq 2\sqrt{E[h(X)^2]E[h'(X)^2]} \\
\sqrt{E[h(X)^2]} &\leq 2\sqrt{E[h'(X)^2]} \\
E[h(X)^2] &\leq 4E[h'(X)^2] \\
Var(g(X)) &= Var(h(X)) \\
&\leq E[h(X)^2] \\
&\leq 4E[h'(X)^2] \\
&= 4E[g'(X)^2]
\end{aligned}$$

Thus, the one dimensional Poincare inequality is $Var(g(X)) \leq 4E[g'(X)^2]$. Now, we expand to n dimensions using the Efron-Stine inequality. Let $Z = f(X_1, \ldots, X_n)$. Fix $i \in \{1, \ldots, n\}$ and let $g_i(x) = f(X_1, \ldots, X_{i-1}, x, X_{i+1}, \ldots, X_n)$. Because f and its first partial derivatives are bounded, g_i is bounded with a bounded derivative. $g_i'(x) = \frac{\partial f}{\partial x_i}(X_1, \ldots, X_{i-1}, x, X_{i+1}, \ldots, X_n)$. Since X_i is independent of X_{-i}, its conditional distribution given X_{-i} remains Exp(1). Therefore, we can apply the one dimensional Poincare inequality:

$$\begin{aligned}
Var^{(i)}(f(X_1, \ldots, X_n)) &= Var(g_i(X_i)|X_{-i}) \\
&\leq 4E[g_i'(X_i)^2|X_{-i}] \\
&= 4E\left[\left(\frac{\partial f}{\partial x_i}(X_1, \ldots, X_n)\right)^2 |X_{-i}\right] \\
E[Var^{(i)}(Z)] &\leq 4E\left[E\left[\left(\frac{\partial f}{\partial x_i}(X_1, \ldots, X_n)\right)^2 |X_{-i}\right]\right] \\
&= 4E\left[\left(\frac{\partial f}{\partial x_i}(X)\right)^2\right]
\end{aligned}$$

Now, use Efron-Stein:

$$\begin{aligned}
Var(f(X)) &\leq \sum_{i=1}^{n} E[Var^{(i)}(Z)] \\
&\leq 4\sum_{i=1}^{n} E\left[\left(\frac{\partial f}{\partial x_i}(X)\right)^2\right] \\
&= 4E\left[\sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}(X)\right)^2\right]
\end{aligned}$$

By the definition of the squared Euclidean norm of the gradient, $\|\nabla f(X)\|_2^2 = \sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}(X)\right)^2$, thus $Var f(X_1, \ldots, X_n) \leq 4E[\|\nabla f(X_1, \ldots, X_n)\|_2^2]$, which is the n-dimensional Poincare inequality.